LENGTH DIFFERENCE FOR THE IDEAL OF DENOMINATORS OVER A ONE DIMENSIONAL NOETHERIAN LOCAL RING WITHOUT EMBEDDED PRIMES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02QH (sources/stacks-chow/pinned-original-source.tex line 18594).

Allowed standard prerequisites (editorial).

- Additivity of divisors for the restricted situation with a and b. Same chapter, Lemma labelled lemma-additivity-divisors-restricted (source line 373), invoked at proof lines 18641-18643. Supplies the formula that turns length_A(M/aM) - length_A(M/bM) into the sum over the minimal primes; the reduction to that difference is carried out in the counted span.
- The order function ord_{A/q} on the residue field of a one dimensional quotient. Algebra, Definition labelled algebra-definition-ord (Stacks tag 02MD), invoked at proof line 18644. Fixes the meaning of the ord-functions appearing on the right hand side.

Source reference. Original lemma 02QG: sources/stacks-chow/pinned-original-source.tex, source lines 373–373. Same-chapter pointer into the author environment lemma at source line 373; invoked at proof line 18642 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 02MD: sources/stacks-chow/tags.txt, source lines 3440–3440. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 18644 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let A be a Noetherian local ring. Let M be a finite A-module. Let $a, b \in A$. Assume*

(1) $\dim(A) = 1$,
(2) *both a and b are nonzerodivisors in A,*

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(3) *A has no embedded primes,*
(4) *M has no embedded associated primes,*
(5) *$Supp(M) = \mathrm{Spec}(A)$.*

Let $I = \{x \in A \mid x(a/b) \in A\}$. Let $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$ be the minimal primes of A. Then $(a/b)IM \subset M$ and

$$length_A(M/(a/b)IM) - length_A(M/IM) = \sum\nolimits_i length_{A_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i}) ord_{A/\mathfrak{q}_i}(a/b)$$

Proof. Since M has no embedded associated primes, and since the support of M is $\mathrm{Spec}(A)$ we see that $\mathrm{Ass}(M) = \{\mathfrak{q}_1, \ldots, \mathfrak{q}_t\}$. Hence a, b are nonzerodivisors on M. Note that

$$\begin{aligned}
&\mathrm{length}_A(M/(a/b)IM) \\
&= \mathrm{length}_A(bM/aIM) \\
&= \mathrm{length}_A(M/aIM) - \mathrm{length}_A(M/bM) \\
&= \mathrm{length}_A(M/aM) + \mathrm{length}_A(aM/aIM) - \mathrm{length}_A(M/bM) \\
&= \mathrm{length}_A(M/aM) + \mathrm{length}_A(M/IM) - \mathrm{length}_A(M/bM)
\end{aligned}$$

as the injective map $b : M \to bM$ maps $(a/b)IM$ to aIM and the injective map $a : M \to aM$ maps IM to aIM. Hence the left hand side of the equation of the lemma is equal to

$$\mathrm{length}_A(M/aM) - \mathrm{length}_A(M/bM).$$

Applying the second formula of Lemma 02QG with $x = a, b$ respectively and using Algebra, Definition 02MD of the ord-functions we get the result. □